

Guidance Theory

Gergely Attila

December 16, 2025

Optimal Constant Heading for Shortest Time Path

Problem Statement

Given:

- Initial position (λ_0, φ_0) and target (λ_1, φ_1) (nearby points)
- Constant speed v (m/s)
- Instantaneous heading control $\psi(t)$ (no yaw-rate constraint)
- Local validity of kinematic equations

Find: heading ψ^* that minimizes travel time.

Kinematic Model (WGS-84)

$$\dot{\varphi} = \frac{v \cos \psi}{M(\varphi)}, \quad \dot{\lambda} = \frac{v \sin \psi}{N(\varphi) \cos \varphi}$$

where $M(\varphi)$ and $N(\varphi)$ are the meridian and prime vertical radii of curvature.

Local Linearization

For nearby points, introduce local Cartesian coordinates around (λ_0, φ_0) :

$$x = N(\varphi_0) \cos \varphi_0 \cdot (\lambda - \lambda_0), \quad y = M(\varphi_0) \cdot (\varphi - \varphi_0)$$

The dynamics become:

$$\dot{x} = v \sin \psi, \quad \dot{y} = v \cos \psi$$

This is exactly a **planar system** with constant speed and free heading.

Optimality Condition

With constant speed v , minimizing path length is equivalent to minimizing time:

$$\min L = \int_0^{t_f} v dt \iff \min t_f$$

In the plane with unrestricted heading, the shortest path is a **straight line**.

Optimal Heading

Define target displacement in local coordinates:

$$\Delta x = N(\varphi_0) \cos \varphi_0 \cdot (\lambda_1 - \lambda_0)$$

$$\Delta y = M(\varphi_0) \cdot (\varphi_1 - \varphi_0)$$

The optimal constant heading is:

$$\boxed{\psi^* = \text{atan2}(\Delta x, \Delta y)}$$

or equivalently:

$$\boxed{\psi^* = \text{atan2} \left(N(\varphi_0) \cos \varphi_0 (\lambda_1 - \lambda_0), M(\varphi_0) (\varphi_1 - \varphi_0) \right)}$$

Properties

- ψ^* is the **initial bearing** along the local geodesic
- The heading is **constant**: $\dot{\psi} = 0$
- Any time-varying $\psi(t)$ results in a longer path
- This solution assumes **instantaneous heading control** (no yaw-rate limit)

Validity Range

This solution is accurate when:

- Points are close enough for linear approximation
- $M(\varphi)$ and $N(\varphi)$ can be treated as constant
- For distant points, use great-circle initial bearing instead